**Answer ALL Questions**

**Question 1 and 2 carries 25 marks each, Question 3 carries 30 marks**

1. a) There are five isolated AC systems (A, B, C, D and E) each with a rated voltage of 400 kV as shown in Figure 1.

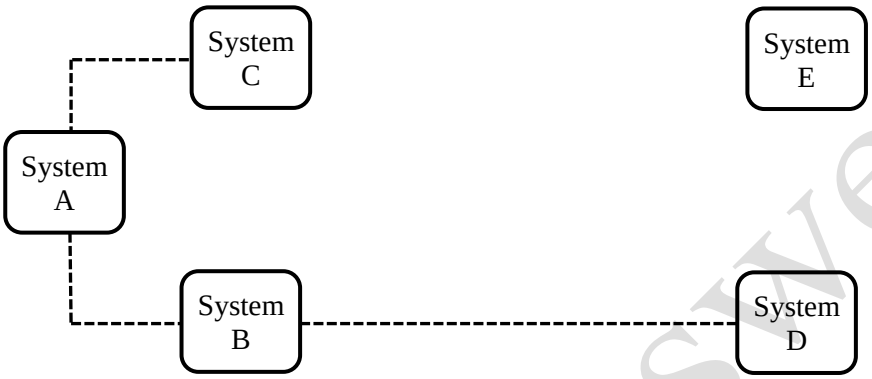



Figure 1.1: Five isolated 400 kV AC systems with planned interconnections shown by dotted lines

The short-circuit level (SCL) at a particular bus within each of these systems where interconnections are planned are provided in Table 1.

| System | A | B | C | D | E |
|---|---|---|---|---|---|
| SCL (GVA) | 20 | 30 | 25 | 50 | 55 |

Table 1.1: Short-circuit level at a particular bus for different systems

The plan is to interconnect system B with D via a point-to-point LCC-HVDC link. The reactive power provision at each end of the LCC-HVDC link would be limited to 65% of its rated active power capacity. The minimum effective short circuit ratio (ESCR) is 4.0. Calculate the maximum rated power capacity of the HVDC link connecting B and D under the following situations:

i) Systems A, B are interconnected by a 400 kV AC line with 5 Ω impedance while systems C and E are isolated.

[7]

The equivalent impedances at the bus in each system are:

$$Z_A = \frac{V^2}{S_A} = \frac{400^2}{20 \times 10^3} = 8\ \Omega$$

$$Z_B = \frac{V^2}{S_B} = \frac{400^2}{30 \times 10^3} = 5.33\ \Omega$$

$$Z_C = \frac{V^2}{S_C} = \frac{400^2}{25 \times 10^3} = 6.4\ \Omega$$

Model Answers

$$Z_D = \frac{V^2}{S_D} = \frac{400^2}{50 \times 10^3} = 3.2\ \Omega$$

[2 marks]

With system A and B connected by a 400 kV AC line with an impedance of 5 Ω, and system C isolated, the equivalent impedance at the bus in system B is

$$Z'_B = Z_B \parallel (Z_{AB} + Z_A) = 3.782\ \Omega$$

[1 mark]

Revised SCL at the bus in system B is

$$S'_B = \frac{V^2}{Z'_B} = 42.31\ \text{GVA}$$

[1 mark]

$$\text{ESCR} = \frac{S - Q_c}{P_{DC}} = \frac{S - xP_{DC}}{P_{DC}} \geq \text{ESCR}_{\min} \Rightarrow P_{DC} \leq \frac{S}{x + \text{ESCR}_{\min}}$$

[1 mark]

$$P_{DC-B} \leq \frac{S'_B}{x + \text{ESCR}_{\min}} = 9.098\ \text{GW}$$
$$P_{DC-D} \leq \frac{S_D}{x + \text{ESCR}_{\min}} = 10.753\ \text{GW}$$

$$P_{DC-BD} = \min_P\{P_{DC-B}, P_{DC-D}\} = 9.098\ \text{GW}$$

[2 marks]

ii) Systems A and B are interconnected by a 400 kV AC line with 5 Ω impedance while systems A and C are interconnected by a 400 kV AC line with 7.5 Ω impedance.

[5]

With system A and B connected by a 400 kV AC line with an impedance of 5 Ω, and system A and C connected by a 400 kV AC line with an impedance of 7.5 Ω, the equivalent impedance at the bus in system B is

$$Z''_B = Z_B \parallel [Z_{AB} + \{Z_A \parallel (Z_{AC} + Z_C)\}] = 3.488\ \Omega$$

[2 marks]

Model Answers

Revised SCL at the bus in system B is

$$S_B'' = \frac{V^2}{Z_B''} = 45.877 \text{ GVA}$$

[1 mark]

$$P_{DC-B1} \leq \frac{S_B''}{x + \text{ESCR}_{\min}} = 9.866 \text{ GW}$$

$$P_{DC-BD1} = \min_P\{P_{DC-B1}, P_{DC-D}\} = 9.866 \text{ GW}$$

[2 marks]

b) Consider the standard six-pulse line commutated converter (LCC) configuration with the thyristor valves numbered in the usual order.

i) Assuming ideal thyristor valves, complete Table 1.2 by entering the expressions for the direct voltage ($v_d$) and voltages across thyristor valve 3 ($v_3$) and 4 ($v_4$) when the thyristor valves shown in the first column are conducting. The expressions must be in terms of phase voltage (e.g. $e_a, e_b, e_c$) or line voltage (e.g. $e_{bc}, e_{ca}$ etc.).

| Thyristor valves conducting | $v_d$ | $v_3$ | $v_4$ |
|---|---|---|---|
| 5, 6, 1 | | | |
| 6, 1 | | | |
| 6, 1, 2 | | | |
| 1,2 | | | |

**Table 1.2: Direct voltage and thyristor valve voltage for different thyristors in conduction**

[1×12]

| Thyristor valves conducting | $v_d$ | $v_3$ | $v_4$ |
|---|---|---|---|
| 5, 6, 1 | $-\frac{3}{2}e_b$ | $e_{ba}$ | $\frac{3}{2}e_b$ |
| 6, 1 | $e_{ab}$ | $e_{ba}$ | $e_{ba}$ |
| 6, 1, 2 | $\frac{3}{2}e_a$ | $-\frac{3}{2}e_a$ | $e_{ba}$ |
| 1,2 | $e_{ac}$ | $e_{ca}$ | $e_{ba}$ |

[1 mark for each entry]

Model Answers

ii) A commutation failure occurs while thyristor valves $5, 6, 1$ are conducting. Which thyristor valves on the same leg would conduct next creating a short circuit across the converter? Assume there is no commutation failure on the other row.

[1]

Thyristor valves 5 would continue to conduct instead of 1 turning on and after 2 turns on, 5 and 2 on the same leg would conduct.

Model Answers

2. A 2 GW, ±500 kV point-to-point HVDC link interconnects two separate AC systems. The HVDC link uses a standard 12-pulse converter arrangement for each pole at both ends. At a given time, the AC-side voltage at the rectifier and inverter ends are 402 kV and 393 kV, respectively. The parameters of the converter at each end and the HVDC link are given in Table 2.1:

| | |
|---|---|
| Minimum firing angle | 4° |
| Minimum extinction advance angle | 17° |
| Resistance of each pole | 4.5 Ω |
| Commutation resistance for each 6-pulse converter | 5.0 Ω |
| Current margin | 15% |
| Turns ratio of converter transformer with low voltage on converter side | 0.5 |

**Table 2.1: Parameters of the converter at each end and the HVDC link**

a) Under normal condition, the rectifier end of the HVDC link maintains the rated direct current while the inverter end operates at the minimum extinction advance angle. Calculate the following considering an equivalent monopole configuration:

i) Firing angle (in degrees) at the rectifier.

[5]

Calculation considering an equivalent monopole:

$B = 4, P_{d-rated} = 2000$ MW, $V_{d-rated}$=2×500 kV=1000 kV,

$E_{acr} = 402$ kV, $E_{aci} = 393$ kV, $T = 0.5$

$$I_d = \frac{P_{d-rated}}{V_{d-rated}} = 2 \text{ kA}$$

At inverter end

$$V_{doi} = \frac{3\sqrt{2}}{\pi} E_{aci} BT = 1061.5 \text{ kV}$$

[1 mark]

$$V_{di} = V_{doi} \cos \gamma_{min} - BR_c I_d = 975.09 \text{ kV}$$

[1 mark]

At rectifier end

$$V_{dr} = V_{dr} + 2 \times R_L I_d = 993.09 \text{ kV}$$

[1 mark]

$$V_{dor} = \frac{3\sqrt{2}}{\pi} E_{acr} BT = 1085.8 \text{ kV}$$

Model Answers

[1 mark]

$$\alpha = \cos^{-1}\left[\frac{V_{dr} + BR_c I_d}{V_{dor}}\right] = 17.92°$$

[1 mark]

ii) Total reactive power consumed by the rectifier.

[2]

Power factor angle at rectifier

$$\phi_r = \cos^{-1}\left(\frac{V_{dr}}{V_{dor}}\right) = 23.85°$$

$$Q_r = V_{dr} I_d \tan\phi_r = 877.93 \text{ MVAr}$$

[1 mark each – 2 marks]

iii) Total reactive power consumed by the inverter.

[2]

Power factor angle at inverter

$$\phi_i = \cos^{-1}\left(\frac{V_{di}}{V_{doi}}\right) = 23.27°$$

$$Q_i = V_{di} I_d \tan\phi_i = 838.86 \text{ MVAr}$$

[1 mark each – 2 marks]

iv) Commutation overlap angle (in degrees) at the inverter.

[2]

$$V_{di} = \frac{V_{doi}}{2}[\cos\gamma + \cos(\gamma + \mu)]$$

$$\mu_i = \cos^{-1}\left(\frac{2V_{di}}{V_{doi}} - \cos\gamma_{min}\right) - \gamma_{min} = 32.85°$$

[1 mark each – 2 marks]

b) If the inverter end AC voltage is reduced by 10%, calculate the following considering an equivalent monopole configuration:

i) Firing angle (in degrees) at the rectifier and extinction advance angle (in degrees) at the inverter.

[4]

Model Answers

The control mode at the two ends remains the same after reduction of inverter end voltage i.e. rectifier continues to control current while the inverter end operates at minimum extinction angle

$\gamma = \gamma_{min} = 17°$

[1 mark]

At inverter end

$E'_{aci} = 0.9 \times E_{aci} = 353.7 \text{ kV}$
$V'_{doi} = \frac{3\sqrt{2}}{\pi} E'_{aci} BT = 955.33 \text{ kV}$

[1 mark]

$V'_{di} = V'_{doi} \cos \gamma_{min} - BR_c I_d = 873.58 \text{ kV}$

[1 mark]

$V'_{dr} = V'_{di} + 2 \times R_L I_d = 891.58 \text{ kV}$
[1 mark]

$\alpha' = \cos^{-1}\left[\frac{V'_{dr} + BR_c I_d}{V_{dor}}\right] = 30.91°$
[1 mark]

ii) Change in total reactive power drawn by the rectifier with respect to the normal condition in part (a).

[3]

Power factor angle at rectifier
$\phi'_r = \cos^{-1}\left(\frac{V'_{dr}}{V_{dor}}\right) = 34.8°$
$Q'_r = V'_{dr} I_d \tan \phi'_r = 1239.4 \text{ MVAr}$

$\Delta Q = Q'_r - Q_r = 361.43 \text{ MVAr}$

[1 mark each – 3 marks]

iii) Commutation overlap angle (in degrees) at the rectifier.

[2]

$V'_{dr} = \frac{V_{dor}}{2}[\cos \alpha' + \cos(\alpha' + \mu')]$

Model Answers

$$\mu_r' = \cos^{-1}\left(\frac{2V_{dr}'}{V_{dor}} - \cos\alpha'\right) - \alpha' = 7.43°$$

[1 mark each – 2 marks]

iv) Effective turns ratio of the converter transformers at each end needed to maintain the rated direct voltage at the inverter end with minimum reactive power consumption at both ends.

[5]

Inverter end

$$V_{di}'' = V_{d-rated} = 1000 \text{ kV}$$

$$V_{doi}'' = \frac{V_{d-rated} + BR_cI_d}{\cos\gamma_{min}} = 1087.5 \text{ kV}$$

[1 mark]

$$T_i = \frac{V_{doi}'' \times \pi}{3\sqrt{2} \times E_{aci}'B} = 0.5692$$

[1 mark]

Rectifier end

$$V_{dr}'' = V_{di}'' + 2 \times R_LI_d = 1018 \text{ kV}$$

[1 mark]

$$V_{dor}'' = \frac{V_{dr}'' + BR_cI_d}{\cos\alpha_{min}} = 1060.6 \text{ kV}$$

[1 mark]

$$T_i = \frac{V_{dor}'' \times \pi}{3\sqrt{2} \times E_{acr}B} = 0.4884$$

[1 mark]

Model Answers

3. a) Answer the following questions related to the DC-DC converter shown in Figure 3.1. The converter parameters are summarised in Table 3.1:

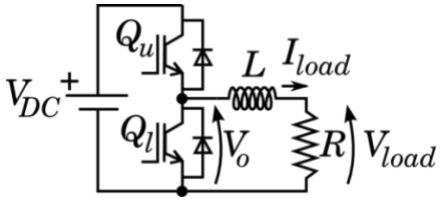



**Figure 3.1 – Diagram of a DC-DC converter feeding a resistive load**

| Parameter | Value | Units | Brief description |
|---|---|---|---|
| $V_{DC}$ | 200 | V | Rated DC link voltage |
| $R$ | 56 | Ω | Load resistor |
| $P_{rated}$ | 400 | W | Rated active power delivered to the load |
| $f_{sw}$ | 150 | kHz | Rated switching frequency |

**Table 3.1 – Characteristic parameters of the DC-DC converter**

(i) Write the linear ordinary differential equation (ode) that relates $V_o(t)$ and $I_{load}(t)$ and its corresponding transfer function in Laplace variables.

[3]

Applying Kirchhoff's voltage law going through the loop formed by the two voltage sources and the filter and adding their voltages together, we obtain:

$$V_o(t) = L\frac{dI_{load}(t)}{dt} + RI_{load}(t)$$

[1.5 marks]

Applying the Laplace transform to this equation and solving for $I_{load}/V_o$ we get:

$$G(s) \triangleq \frac{I_{load}(s)}{V_o(s)} = \frac{1}{Ls + R}$$

[1.5 marks]

(ii) Calculate the duty cycle, $D$, the average of the load current, $I_{load}$, and the average of the current extracted from the DC bus, $I_{DC}$, when delivering the rated power to the load.

[3]

We can start by calculating the voltage the resistor requires in order for its power to match the rated power of the application:

$$V_{load} = \sqrt{P_{rated}R} = 149.7\ V$$

The load current is: $I_{load} = \frac{P_{rated}}{V_{load}} = 2.67\ A$

The resistance of the filter is neglected, therefore the average of the output voltage of the bridge will match the average of the load voltage. We can then calculate the duty cycle as:

$$D = \frac{V_o}{V_{DC}} = \frac{V_{load}}{V_{DC}} = 74.9\%$$

We can calculate the current of the DC bus as: $I_{DC} = I_{load}D = 2\ A$

[3 marks]

Model Answers

(iii) A PI is to be used as as a load current controller. The input of the PI will be the error of the current (ie $e=I_{load}^*-I_{load}$) and the output will be the desired average output voltage $V_o^*$ to be given to the PWM. Calculate the gains of the PI, ie $K_P$ and $K_I$ to achieve zero steady state error and a settling time of 1 ms. **Consider $L$=3 mH.** Justify your answer.

[4]

The transfer function of the system is $\frac{I_{load}}{V_o^*}(s) = \frac{1}{Ls+R}$. One possible design procedure to obtain the desired closed-loop performance is to use an inverse-based design, where the target closed-loop transfer function is of the form $T(s) = \frac{1}{\tau^* s+1}$ with $\tau^*$=0.33 ms (ie the settling time divided by 3). This closed-loop transfer function implies that the open loop transfer function is of the form $GK(s) = \frac{1}{\tau^* s}$. Therefore, the transfer function of the controller is of the form $K(s) = \frac{Ls+R}{\tau^* s}$, which corresponds to a PI transfer function. By operating with this transfer function we can leave it in the form:

$$K(s) = \frac{\frac{L}{\tau^*}s + \frac{R}{\tau^*}}{s} = \frac{K_P s + K_I}{s}$$

This gives: $K_P$=9 V/A, $K_I$= 168 kV/(A·s)

This method is widely used in the module. Students may choose a different solution that isn't based on making the closed-loop transfer function be equivalent to a first order system using the pole-cancelation. This will be considered correct as long as the result meets the specification of zero steady-state error and a settling time of 1 ms.

[4 marks]

b) Answer the following questions related to the AC-DC converter shown in Figure 3.2. The converter parameters are summarised in Table 3.2.:

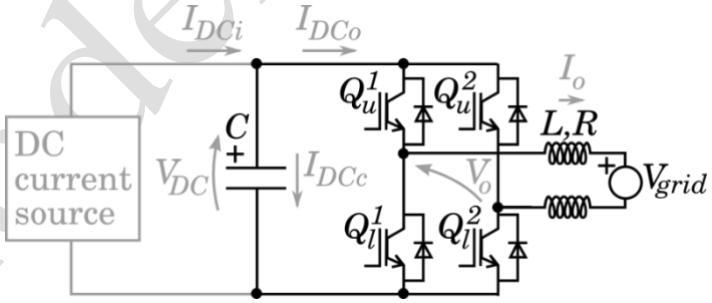



**Figure 3.2 – Diagram of a single phase inverter connected to an AC grid**

| Parameter | Value | Units | Brief description |
|---|---|---|---|
| $V_{grid}$ | 120 | V rms | Rated AC grid voltage |
| $V_{DC}$ | 300 | V | Rated DC link voltage |
| $P_{rated}$ | 1 | kW | Rated active power delivered to the grid |
| $f_{rated}$ | 50 | Hz | Rated frequency of the AC grid |
| $R$ | 0.1 | Ω | Equivalent resistance of bridge plus the filter |
| $L$ | 5 | mH | Inductance of the filter |

**Table 3.2 – Diagram of a single phase inverter connected to an AC grid**

Model Answers

(i) Answer the following questions related to the following operating points:

- For an operating point with zero current at the output of the converter. What is the output voltage of the converter, $V_o$? (use Vgrid phasor as the reference of angle). What is the minimum $V_{DC}$ voltage required to apply $V_o$?

[3]

The output current is zero, therefore the 50 Hz voltage across the output filter is zero. This means that $V_o = V_{grid} = 120$ V with angle zero.
[1 mark]

The bridge can generate voltages, $V_o$, between $+V_{DC}$ and $-V_{DC}$, this range has to be wide enough to reach the peak values of $V_o$, which are $\pm\sqrt{2} \cdot 120$ V. Therefore $V_{DC}$ must be greater than 170 V.
[2 marks]

- For rated output power ($P_{grid}=P_{rated}$, $Q_{grid}=0$). What is the output current? How much reactive power does the converter bridge have to feed to the output filter? What are the approximate conduction losses?

[2]

The output current is $I_{rated} = \frac{P_{rated}}{V_{grid}} = 8.33$ A, therefore the reactive power of the filter is $Q_{filter}=X_{filter}I_{rated}^2=109$ VAr.
[2 mark]

- For a given operating point where the 100 Hz component of $I_{DCo}$ is of 3.5 A of amplitude. What is the minimum capacitance, $C$, the DC bus would need for the 100 Hz ripple of $V_{DC}$ caused by this current to be less than 1V of amplitude?

[2]

The 100 Hz ripple of $I_{DCo}$ is caused by the interaction between the 50 Hz components of $V_o$ and $I_o$. This current flows through $C$ and causes a 100 Hz ripple of $V_{DC}$. If the amplitude of the 100 Hz component of $I_{DCc}$ is 3.5 A and we want it to cause a voltage ripple of $V_{DC}$ that is of 1 V of amplitude, this gives us a boundary of how low the capacitance must be: $X_C \leq 0.286\ \Omega$. Therefore, $C \geq 5.56$ mF.
[2 mark]

(ii) Regarding the design of a DC bus voltage controller for this converter, we know that $V_{DC}$ will have a 100 Hz ripple under normal operation and we can't prevent it. Explain briefly what options we have in order to make the voltage controller ignore the 100 Hz ripple.

[2]

The 100 Hz ripple of $I_{DCo}$ is caused by the interaction between the 50 Hz components of $V_o$ and $I_o$ and it can't be prevented through the control of the converter. If the $V_{DC}$ controller isn't designed to have low gain at this frequency, it may cause the reference of $I_o$ to contain harmonic distortion. This can be done by setting the bandwidth of the VDC controller to be low enough to make the 100 Hz component of $P_{DCo}$* small enough, if for example the controller was a simple PI. Alternatively, the gain can be made to be smaller by adding a pair of conjugate zeros at the frequency of 100 Hz by adding a notch filter in

Model Answers

series with the voltage controller. This has a detrimental effect on the stability of the controller and it has to be tuned properly.
[2 mark]

(iii) Regarding the design of the output current controller for this converter, would the steady state error be zero if we chose to use a PI controller? Provide a brief justification.

[3]

The output current of the converter is a sine wave of 50 Hz. For the steady state error to be zero, the closed-loop transfer function must be stable and it's gain at 50 Hz has to be zero. If a PI controller is used, we get the following open-loop transfer function:

$$GK(s) = \frac{V_o(s)}{err(s)} = \frac{K_p s + K_i}{L\,s^2 + R\,s}$$

and the corresponding closed-loop transfer function becomes:

$$T(s) = \frac{I_o(s)}{I_o^*(s)} = \frac{K_p s + K_i}{L\,s^2 + (K_P + R)s + K_i}$$

This transfer function can't be made to have 0 gain at 50 Hz, therefore the error can't be equal to zero.

Full marks for this question will only be awarded if a brief derivation of the closed-loop equations or some form of formulation is provided to justify the answer.
[2 mark]

c) Consider a basic one-arm modular converter connected between an AC grid and a DC grid represented as equivalent voltage sources as shown in Figure 3.3. Answer the questions below:

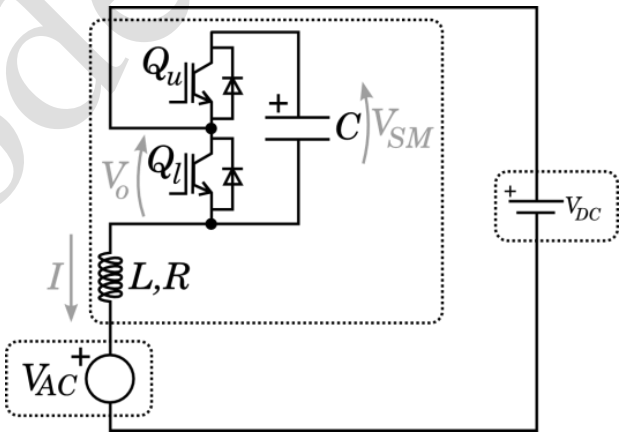



**Figure 3.3 – Diagram of a one-arm modular AC/DC converter**

| Parameter | Value | Units | Brief description |
|---|---|---|---|
| $V_{DC}$ | 500 | V | Rated DC grid voltage. |
| $V_{AC}$ | 230 | V rms | Rated AC grid voltage |
| $f_{rated}$ | 50 | Hz | Rated frequency of the AC gr |
| $R$ | 3 | mΩ | Equivalent resistance of bridge plus the filter |

Model Answers

| | | | |
|---|---|---|---|
| $L$ | 8 | mH | Inductance of the filter |

**Table 3.3 – Diagram of a single phase inverter connected**

(i) What is the output voltage of the arm, $V_o$, when the output current, $I$, is zero? What is the minimum submodule capacitor voltage, $V_{SM}$, required to generate this voltage?

[4]

Applying Kirchoff's laws, it is easy to find that when the current is zero, the following condition must be met: $V_{AC} + V_o = V_{DC}$. $V_{AC}$ is a sine wave of 50 Hz whereas $V_{DC}$ is a constant DC value. Consequently, Vo must contain a DC component of $V_o^{DC}$=500V and a 50 Hz sine wave component of opposite sign to VAC of $V_o^{AC}$=230 V rms.
[2 mark]

The output voltage waveform will be: $V_o(t) = V_o^{DC} + \sqrt{2}V_o^{AC}\sin(2\pi 50t)$. The maximum Vo the converter will apply is $V_o^{DC} + \sqrt{2}V_o^{AC}$, which gives 825 V. The arm can generate voltages of up to $V_{SM}$, therefore $V_{SM}$ must be greater than 825 V.
[2 mark]

(ii) This converter uses a half-bridge submodule, explain how this affects the minimum $V_{DC}$ required for the converter to operate for a given rated $V_{AC}$. Could this limitation be overcome by using a different submodule type?

[4]

The half-bridge submodule can only output positive voltages, this means that if $V_o^{DC} - \sqrt{2}V_o^{AC} < 0$, the converter would be unable to generate the required voltage and wouldn't be able to operate. This can happen of $V_{DC}$ is too low, for example, when a fault happens in the $V_{DC}$ system and it forces the converter to be disconnected, typically interrupting the fault current using a circuit breaker on the AC side of the converter.
[3 mark]

The problem can be overcome by using a submodule topology that is capable of generating negative output voltages. The most common example is the full-bridge submodule.
[1 mark]

Model Answers